

Categories, Proofs, and Processes: Assignment 6

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Question 1

Claim 1.1. *h is monotone.*

Proof. Suppose $x, y \in P, x \leq y$. Then

$$\begin{aligned} h(x) &\leq \bigvee \{h(x), h(y)\} \quad (\text{by LUB}) \\ &= h\left(\bigvee \{x, y\}\right) \\ &= h(y) \quad (\text{since } y = \bigvee \{x, y\}) \end{aligned} \tag{1}$$

So h is monotone. □

Proposition 1.2. *U has a left adjoint.*

Proof. Define the functor $F : \mathbf{Set} \rightarrow \mathbf{SL}$

$$F(X) = (\mathcal{P}(X), \subseteq)$$

$$F(f : X \rightarrow Y) : \mathcal{P}(X) \rightarrow \mathcal{P}(Y) := A \mapsto \{f(a) | a \in A\}$$

which is the covariant powerset functor for arrows, so functoriality is satisfied.

Note that $\bigvee S = \bigvee \{A | A \in S\} = \bigvee_{A \in S} A$. The functor on arrows preserves least upper bounds where $\forall f : X \rightarrow Y$ in $\mathbf{Set}$,

$$\begin{aligned} F(f)\left(\bigvee S\right) &= F(f)\left(\bigcup_{A \in S} A\right) \\ &= \left\{f(a) \mid a \in \bigcup_{A \in S} A\right\} \\ &= \bigcup_{A \in S} \{f(a) | a \in A\} \\ &= \bigcup_{A \in S} F(f)(A) \\ &= \bigvee_{A \in S} F(f)(A) \\ &= \bigvee \{F(f)(A) | A \in S\} \end{aligned} \tag{2}$$

Claim 1.3. $F : \mathbf{Set} \rightarrow \mathbf{SL}$ as defined is the left adjoint of U .

F is the left adjoint of U , iff $\forall A \in \text{Ob}(\mathbf{Set})$, there exists a unit $\eta_A : A \rightarrow UFA$ such that

$$\forall f : A \rightarrow B, \exists! \hat{f} : \mathcal{P}(A) \rightarrow B$$

where $f \in \text{Ar}(\mathbf{Set})$, $\hat{f} \in \text{Ar}(\mathbf{SL})$, the following diagram commutes

$$\begin{array}{ccc} A & \xrightarrow{\eta_A} & \mathcal{P}(A) \\ & \searrow f & \downarrow U(\hat{f}) \\ & & B \end{array}$$

For each object $C \in \text{Ob}(\mathbf{Set})$, define the unit arrow $\eta_C : C \rightarrow UFC$ where $UFC = \mathcal{P}(C)$

$$\eta_C(c) = \{c\}$$

From an arrow $f : A \rightarrow B$ in $\mathbf{Set}$, we can construct an arrow in $\mathbf{SL}$

$$\hat{f} : \mathcal{P}(A) \rightarrow B := X \mapsto \bigvee F(f)(X)$$

where $U(\hat{f}) = \hat{f}$. Then $\forall a \in A$,

$$(U(\hat{f}) \circ \eta_A)(a) = \hat{f}(\{a\}) = f(a)$$

So $f = U(\hat{f}) \circ \eta_A$.

Suppose there's another $g : \mathcal{P}(A) \rightarrow B$ such that $f = U(g) \circ \eta_A = g \circ \eta_A$. Then it also preserves least upper bounds where $\forall X \subseteq A$,

$$\begin{aligned} g(X) &= g\left(\bigvee \{\{a\} | a \in X\}\right) \\ &= \bigvee \{g(\{a\}) | a \in X\} \quad (\text{by def. of arrow in SL}) \\ &= \bigvee \{g \circ \eta_A(a) | a \in X\} \quad (\text{since } \eta_A(a) = \{a\}) \\ &= \bigvee \{f(a) | a \in X\} \\ &= \bigvee F(f)(X) \\ &= \hat{f}(X) \end{aligned} \tag{3}$$

so $g = \hat{f}$ is unique. Thus, F is the left adjoint of U . □

Question 2

Proposition 2.4. *The contravariant powerset functor $\mathcal{P} : \mathbf{Set}^{op} \rightarrow \mathbf{Set}$ is its own left adjoint.*

Proof. $\mathcal{P}^{op}$ is the left adjoint of $\mathcal{P}$ iff $\forall X \in \mathbf{Ob}(\mathbf{Set})$, there exists a unit

$$\eta_X : X \rightarrow \mathcal{P}\mathcal{P}(X)$$

such that $\forall f : X \rightarrow \mathcal{P}(Y)$ in $\mathbf{Set}$, $\exists! \hat{f} : \mathcal{P}^{op}(X) \rightarrow Y$ in $\mathbf{Set}^{op}$ where the diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & \mathcal{P}\mathcal{P}(X) \\ & \searrow f & \downarrow \mathcal{P}(\hat{f}) \\ & & \mathcal{P}(Y) \end{array}$$

Define $\forall X \in \mathbf{Ob}(\mathbf{Set})$, $\eta_X(x) = \{\{x\}\}$.

Define

$$\hat{f}(y) = \{x | y \in f(x)\}$$

in $\mathbf{Set}$, with its corresponding arrow in $\mathbf{Set}^{op}$. Then

$$\mathcal{P}(\hat{f})(A) = \{y | \exists S \in A, \forall x \in S, y \in f(x)\}$$

So

$$\begin{aligned} \mathcal{P}(\hat{f}) \circ \eta_X(x) &= \mathcal{P}(\hat{f})(\{\{x\}\}) \\ &= \{y | y \in f(x)\} \\ &= f(x) \end{aligned} \tag{4}$$

Let $g : \mathcal{P}^{op}(X) \rightarrow Y$ in $\mathbf{Set}^{op}$ such that $\mathcal{P}(g) \circ \eta_X = f$. Then $\forall x \in X$,

$$\mathcal{P}(g)(\{\{x\}\}) = \{g(\{x\})\} = f(x) \subseteq Y$$

Since $g : Y \rightarrow \mathcal{P}^{op}(X)$ in $\mathbf{Set}$ is a function, it must be that $\forall y \in f(x), x \in g(y)$, so

$$g(y) = \{x | y \in f(x)\}$$

$\hat{f} = g$ is the unique morphism.

□

Question 3

A morphism of graphs $f : (V, E, s, t) \rightarrow (V', E', s', t')$ consists of mappings of

- vertices, $f_V : V \rightarrow V'$
- edges, $f_E : E \rightarrow E'$
- source, $f_s(s) = s' \iff (\forall e \in E, s(e) = v \implies s'(f_E(e)) = f_V(v))$
- target, $f_t(t) = t' \iff (\forall e \in E, t(e) = v \implies t'(f_E(e)) = f_V(v))$

Define the functor $F : \mathbf{Grph} \rightarrow \mathbf{Cat}$ with the object map

$$F :: (V, E, s, t) \mapsto \mathcal{C}$$

where the category $\mathcal{C}$ has objects $\text{Ob}(\mathcal{C}) = V$, and arrows $\text{Ar}(\mathcal{C})$ are freely generated from the composable edges E , so $E \subseteq \text{Ar}(\mathcal{C})$, and if $e_1, e_2 \in E$ are such that $s(e_1) = t(e_2)$, then $e_1 \circ e_2 \in \text{Ar}(\mathcal{C})$.

The composition map is clearly defined from the free generation. The identity map 1_u for each object u is included in the free generation, from the empty sequence of edges.

Associativity is satisfied since free generation is associative. Unit axiom is satisfied since the identity map satisfies the properties of the identity.

F then maps graph homomorphisms in $\mathbf{Grph}$ to functors between the corresponding categories in $\mathbf{Cat}$. Let $f : A \rightarrow B$ be in $\text{Ar}(\mathbf{Grph})$.

The object mapping of Ff is clear

$$Ff : F(A) \rightarrow F(B) := C_v \mapsto D_{f_v(v)}$$

where $C_v, D_{f_v(v)}$ are the objects in the categories $F(A), F(B)$ representing the vertices $v, f_v(v)$ in the graphs A, B respectively.

Ff maps an arrow $g : C_u \rightarrow C_v$ in the category $F(A)$ to the corresponding arrow $Ff(g) : D_{f_v(u)} \rightarrow D_{f_v(v)}$ in $F(B)$. Ff also maps identity arrows to identity arrows, which exist due to the free generation.

The identity axiom is clearly satisfied.

Composition is preserved since free generation preserves it.

$$Ff(g \circ h) = Ff(g) \circ Ff(h)$$

Functoriality is satisfied.

F is the left adjoint of U , iff $\forall A \in \text{Ob}(\mathbf{Grph})$, there exists a unit $\eta_A : A \rightarrow UFA$ such that

$$\forall f : A \rightarrow U(B), \exists ! \hat{f} : F(A) \rightarrow B$$

where $f \in \text{Ar}(\mathbf{Grph})$, $\hat{f} \in \text{Ar}(\mathbf{Cat})$, the following diagram commutes

$$\begin{array}{ccc} A & \xrightarrow{\eta_A} & UF(A) \\ & \searrow f & \downarrow U(\hat{f}) \\ & & U(B) \end{array}$$

For each object $C = (V, E, s, t) \in \text{Ob}(\mathbf{Grph})$, define the unit arrow $\eta_C : C \rightarrow UFC$ where $\eta_{C,V} = 1_V, \eta_{C,E} = \iota_E$ - the identity map on vertices, and the inclusion map on edges.

Then for each $f : A \rightarrow U(B)$, define the arrow $\hat{f} : F(A) \rightarrow B$ where

$$\hat{f}(C) = Ff_V(C_V)$$

$$\hat{f}(g : C \rightarrow D) = Ff_E(g_E)$$

with C_V as the underlying vertex representing the object C , and g_E as the underlying edge representing the arrow g . The functor $\hat{f}$ satisfies functoriality, since F does.

Then for any object $C \in \text{Ob}(\mathbf{Grph})$,

$$U(\hat{f}) \circ \eta_A(C) = U(\hat{f})(C) = UFf(C) = f(C)$$

$\hat{f}$ is the unique functor.

Question 4

4.1 (a)

Proposition 4.5. *If f has a right adjoint, then f preserves least upper bounds.*

Proof. Suppose f has a right adjoint $g : Q \rightarrow P$.

$$\begin{aligned} \forall x \in S \subseteq P, x &\leq \bigvee S \quad (\text{by upper bound}) \\ \implies \forall x \in S, f(x) &\leq f\left(\bigvee S\right) \quad (f \text{ is monotone}) \end{aligned} \tag{5}$$

$$\begin{aligned} \implies \bigvee \{f(x) | x \in S\} &\leq f\left(\bigvee S\right) \quad (\text{by least UB}) \\ \forall x \in S \subseteq P, f(x) &\leq \bigvee \{f(x) | x \in S\} \quad (\text{by upper bound}) \\ \implies \forall x \in S, x &\leq g\left(\bigvee \{f(x) | x \in S\}\right) \quad (g \text{ is right adjoint}) \\ \implies \bigvee S &\leq g\left(\bigvee \{f(x) | x \in S\}\right) \quad (\text{by least UB}) \\ \implies f\left(\bigvee S\right) &\leq \bigvee \{f(x) | x \in S\} \quad (f \text{ is left adjoint}) \end{aligned} \tag{6}$$

Then since Q is a poset, these imply that

$$f\left(\bigvee S\right) = \bigvee \{f(x) | x \in S\}$$

□

4.2 (b)

Proposition 4.6. *If f preserves least upper bounds, then it has a right adjoint g given by*

$$g(y) = \bigvee \{x \in P | f(x) \leq y\}$$

Proof. Suppose f preserves least upper bounds. We want to show that

$$\forall x \in P, z \in Q, x \leq g(z) \iff f(x) \leq z$$

($\implies$) Let $x \in P, z \in Q$ such that $x \leq g(z)$.

$$\begin{aligned} x \leq g(z) &\iff x \leq \bigvee \{x \in P | f(x) \leq z\} \\ \implies f(x) &\leq f\left(\bigvee \{x \in P | f(x) \leq z\}\right) \\ &\iff f(x) \leq \bigvee \{f(x) | f(x) \leq z\} \quad (f \text{ preserves LUB}) \\ \implies f(x) &\leq z \quad (\text{by least UB}) \end{aligned} \tag{7}$$

($\impliedby$) Now let $f(x) \leq z$. Note that by LUB, $x \in S \implies x \leq \bigvee S$.

$$\begin{aligned} f(x) \leq z &\implies g \circ f(x) \leq g(z) \\ &\iff \bigvee \{x' \in P | f(x') \leq f(x)\} \leq \bigvee \{x \in P | f(x) \leq z\} \\ \implies x &\leq \bigvee \{x \in P | f(x) \leq z\} \quad (f(x) \leq f(x) \text{ by reflexivity}) \\ &\iff x \leq g(z) \end{aligned} \tag{8}$$

□

4.3 (c)

Proposition 4.7. *Complete semi-lattices P also have infima (greatest lower bounds) $\bigwedge S$ for all subsets $S \subseteq P$.*

Proof. Note that the bottom $\bigvee \emptyset = \perp \in P$ exists, where $\forall x \in P, \perp \leq x$, from the existence of LUB for $\emptyset$, and since $\forall x \in P, x' \in \emptyset, x' \leq x$ is vacuously true.

Let $S \subseteq P$ be a subset, with $B \subseteq P$ as the set of lower bounds for S . Since $\perp \in B, B \neq \emptyset$. Take $\bigvee B$, which exists.

Claim 4.8.

$$\bigvee B = \bigwedge S$$

Note that just from definitions,

$$\forall x \in B, x \leq \bigvee B$$

$$\forall x \in B, y \in S, x \leq y$$

So $\forall y \in S, y$ is an upper bound of B . Then by the *least* UB of set B , we must have

$$\forall y \in S, \bigvee B \leq y$$

So $\bigwedge S = \bigvee B \in B$ is the GLB of S .

□

4.4 (d)

From (a) and (b), we have the following

Corollary 4.9. *$f : P \rightarrow Q$ has a right adjoint $g : Q \rightarrow P$ iff f preserves least upper bounds, where*

$$g(y) = \bigvee \{x \in P \mid f(x) \leq y\}$$

Proposition 4.10. *If g has a left adjoint, then g preserves greatest lower bounds.*

Proof. Suppose g has a right adjoint $f : P \rightarrow Q$.

$$\begin{aligned} \forall y \in S \subseteq Q, \bigwedge S &\leq y \quad (\text{by lower bound}) \\ \implies \forall y \in S, g\left(\bigwedge S\right) &\leq g(y) \quad (g \text{ is monotone}) \\ \implies g\left(\bigwedge S\right) &\leq \bigwedge \{g(y) \mid y \in S\} \quad (\text{by greatest LB}) \end{aligned} \tag{9}$$

$$\begin{aligned} \forall y \in S \subseteq Q, \bigwedge \{g(y) \mid y \in S\} &\leq g(y) \quad (\text{by lower bound}) \\ \implies \forall y \in S, f\left(\bigwedge \{g(y) \mid y \in S\}\right) &\leq y \quad (f \text{ is right adjoint}) \\ \implies f\left(\bigwedge \{g(y) \mid y \in S\}\right) &\leq \bigwedge S \quad (\text{by least UB}) \\ \implies \bigwedge \{g(y) \mid y \in S\} &\leq g\left(\bigwedge S\right) \quad (g \text{ is left adjoint}) \end{aligned} \tag{10}$$

Then

$$g\left(\bigwedge S\right) = \bigwedge \{g(y) \mid y \in S\}$$

g preserves greatest lower bounds.

□

Dualising, we get

Corollary 4.11. *$g : Q \rightarrow P$ has a left adjoint iff g preserves greatest lower bounds.*